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The Dependency of Wage Contracts on Monetary Policy

by

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In response to the new-classical argument that anticipated monetary policy is ineffective, one New-Keynesian rebuttal posits sticky wages stemming from long-term labor contracts. We argue here that such fixed-wage contracts may reflect stabilization policy itself. Using an illustrative quantitative model, implemented by a computer program that computes the game payoffs, we analyze wage-contract strategies that could arise in response to various monetary policy strategies. (JEL: E 24, E 58).

1. Introduction

An important principle of new-classical economics is the "invariance proposition," which states that "any predictable part of the money supply should have *no effect* on output, employment, or any other real variables in the economy" (SHEFFRIN [1983, 40]). FISCHER [1977, 194], arguing against this proposition, observes that "the use of longer-term nominal contracts puts an element of stickiness into the nominal wage," causing even fully anticipated monetary policy to have real effects. Because long-term fixed-wage contracts make contracting agents vulnerable to supply and demand shocks during the contracts' duration, Fischer's argument suggest that monetary policy may be desirable as well as non-neutral.

Fischer does not attempt to explain why long-term fixed-wage contracts exist in the first place; he mentions transaction costs as part of a possible explanation. But transactions cost savings would rationally be set against the anticipated costs associated with wage stickiness.

PARKIN [1986, 200] notes that price and wage stickiness cannot be considered exogenous, given rational expectations, but "result from the monetary policy process." MCCALLUM [1980, 730] has shown how prices may be temporarily fixed and yet "reflect expectations formed previously about the current-period money stock." Such statements imply that long-term fixed-wage contracts may be a rational-expectations-based response to certain types of monetary policy. This was also seen by PHELPS [1978, 209], who cites the possibility that "wage setters expect the central bank to *accommodate* [shocks] by adjusting the money supply and thus the price level..."

Thus a monetary policy which seeks to avoid monetary shocks may also allow the survival of sticky-wage contracts. If the monetary authority behaved differently, e.g. either by attempting to exploit the Phillips curve or by keeping the money supply constant (as if the authority were indifferent to fluctuations in employment due to shocks), then long-term fixed-wage contracts might dissolve. One might instead observe contracts with cost-of-living clauses or other responses to anticipated price changes. The persistence of sticky-wage contracts will, in short, reflect the adaptation of the contracting agents to a particular monetary policy regime.

Despite some awareness of this possibility, the view persists that long-term contracts of the Fischer type are a source of nominal rigidity that, by implication, provide a rationale for monetary accommodation (GORDON [1990]). This view fails to consider how "contractual arrangements would alter if the authorities attempted to make use of ... apparent rigidities ..." (KANTOR [1979, 1437]). If contractors react to the monetary authority's strategy, and the authority reacts to what it expects the contractors to do, "a gaming type problem would emerge." As BUITER [1980, 42] has noted, a "lot of work remains to be done" in showing how multiperiod contracts can be generated, possibly as the equilibrium outcome of monetary policy game. This paper, using an illustrative-quantitative model as an example, presents a contribution to such work in the form of an analytical model of a monetary/wage-policy game.

A secondary contribution of this paper is its transformation of the analytical model into a numerical simulation implemented by a computer program. Gaming payoffs are derived by the simulation rather than assigned *ad hoc*. Because the simulation relies upon assumed parameter values, its results do not test the model, but merely serve to illustrate it and to show how robust results derived from it are to changes in particular parameters.

2. A Fischer-Type Macroeconomic Model

The macroeconomic model used in this paper is adapted from the rational expectations model formulated by FISCHER [1977], which, as he states, is similar in spirit to models constructed by SARGENT and WALLACE [1975]. The economy is stationary and closed to foreign trade, with zero growth in the capital stock, but is subjected to random shocks.

Fischer sets up models first for one period and then for two periods. Here we employ a multi-period model using one-period equations that iterate over time, so that both contract length and the flexibility of the nominal wage are outcomes of the model. As GORDON [1990, 1160] notes, contract length is a function of negotiating costs and the "allocative costs of infrequent adjustment." Once a wage contract is negotiated, output becomes demand-determined.

In Fischer's model, during period t half the firms are operating in the first year of a labor contract drawn up at the end of (t-1) and half in the second year of a contract drawn up at the end of (t-2). A simple version of Fischer's supply equation, with variables in logarithm, is

$$y_t^s = \varepsilon(p_t - w_t) + \sigma_t , \tag{1}$$

where y_t^s is the level of output for period t, p_t is the price level for period t, w_t is the wage set at the end of period $t - 1$ for period t, and σ_t is a stochastic real disturbance or supply shock, which allows for changes in the conditions of aggregate supply. The supply of output is thus a decreasing function of the real wage. In our simulation the coefficient ε is assigned a value of unity.

Fischer's demand equation is a simple velocity formula

$$y_t^d = m_t - p_t - \delta_t , \tag{2}$$

where m_t is the logarithm of the money stock during period t, and δ_t is a demand-disturbance term. As FISCHER [1977, 200] notes, δ is a nominal disturbance that, when positive, reduces the price level; hence the negative sign.

Following Fischer, the demand disturbance is a first-order autoregressive equation, where η_t is a disturbance term:

$$\delta_t = \rho\delta_{t-1} + \eta_t , |\rho| < 1 . \tag{3}$$

Nominal demand shocks derive from changes in the demand for money. Random fluctuations in the ratio of money balances to income or to real assets create random fluctuations in aggregate demand. A positive demand shock, i.e. an increase in the aggregate demand for goods, increasing y, would correspond to a negative value of δ.

Our model follows Fischer's framework but extends and alters it so that contracts and wages as well as output are endogenous functions of monetary policy. For the sake of simplicity we confine our analysis to money-demand shocks, abstracting from supply shocks by setting $\sigma_t = 0$ in (1). In sections 3 through 6, we construct the macroeconomic model on which gaming strategies are based. A detailed presentation of the numerical model, with illustrative parameters, is presented in the Appendix. The numerical model is implemented by a computer program, available from the authors. A run of the program yields payoff values for both contracting agents (labor/management) and the monetary authority. The example shows how the results of strategies can be derived as outcomes of the model.

3. Monetary Policy Strategies

Consider three possible monetary policy strategies. A "rigid" or money-targeting policy fixes the money stock independently of the level of prices or employment, e.g. by letting it grow at some fixed (positive or zero) rate. The term "passive" is used to refer to such a policy by WOGLOM [1979, 91]: "A passive rule implies that instrumental variables are unaltered by new information." An "accommodative" or price-level targeting policy adjusts the money stock to insulate the (equilibrium) price level from money demand (velocity) disturbances.[1] Lastly, a "manipulative" policy uses money "surprises" to achieve short-run gains in employment and output in excess of those consistent with stable aggregate demand (i.e. in excess of "natural rates"), as in models of the political business cycle. To achieve control over output, a manipulative policy would also accommodate demand shocks.

The potential gains to the monetary authority from any particular policy include gains associated with output effects of policy and gains associated with non-output related goals. Policy costs include those of monitoring the economy and those of implementing policy.

The objectives of policy makers have been treated in the literature as ranging from broad social welfare to rent seeking.[2] In the present paper, monetary policy strategies vis-a-vis contracting agents will be analyzed on the assumption that the monetary authority (whether acting on its own behalf or as the agent of the government) obtains utility from raising the price level beyond the level expected by the contracting agents, thereby increasing output. (Since benefits to government from a reduction in real debt may compensate for welfare losses caused by inflation, we make no assumptions about the utility to the policy-maker of inflation per se.)

The monetary authority's objective function for a given time period can be written as

$$G = g(y - y_r) - d(w - p) + u - c, \tag{4}$$

where G is the utility expected by the authority. As explained below, we have two variables for output, one (y) for the actual output generated and another (y_r) for what y would be with rigid policy (the explicit implementation of (4) is described in section 5) The real-wage deviation loss d as a function of the nominal wage w and price level p is described in section 4 below, as is the contracting cost c.

Direct utility (u) is associated with an accommodative policy. Policies that benefit contracting agents are assumed for this reason to have some positive political payoff to the monetary authority.

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[1] Our "accommodative" policy is *not* equivalent to an interest-rate targeting policy, in so far as it is aimed at offsetting shocks to *money*, but not necessarily *credit*, demand. See BRUNNER and MELTZER [1976].

[2] See the writings of the "public choice" school, e.g., GWARTNEY and WAGNER [1988].

Fischers' monetary rule, a form of active monetary policy, responds to the disturbance terms σ and δ in (1) and (2):

$$m_t^a = \alpha \sigma_{t-1} + \beta \delta_{t-1} \,. \tag{5}$$

Parameters α and β can, as Fischer related, be set so as to minimize the variance of output by exploiting a fixity of contracts during period 2. Because we are abstracting from supply shocks, the first term in (5) is eliminated. Allowing for this change, we use (5) to represent an accommodative monetary policy, with $\beta = \rho$, as derived by FISCHER [1977, 200], while letting $m^r = 0$ for a rigid policy. Because a manipulative policy encompasses accommodation and adds to it, such a policy is derived from (5):

$$m_t^m = m_t^a + \gamma(w_{t-1}) \,, \tag{6}$$

where the second term is a function of the wage set in the previous period: manipulative policy increases output by increasing the money supply and thereby increasing the price level relative to the wage level. Of course we do not claim this to be the *only* possible form that "manipulative" policy might take; we only wish to illustrate one possibility.

As in Fischer's model, the price level p is a function of the money supply and velocity shocks (which may be offset by monetary policy). Following Fischer, we let the price level adjust each period, but in our model, the price level is normalized to equal the money supply (which in the accommodative case offsets all past disturbances) minus the accumulated previous demand disturbances δ_p:

$$p_t = m_t - \delta_p \,. \tag{7}$$

A "natural-rate" equilibrium real wage of unity is assumed. The nominal wage w is therefore set equal to the price level p at the beginning of all wage contracts and in each period for variable-wage contracts. In fixed-wage contracts, the real wage $(w - p)$ varies as p varies due to demand disturbances.

The final element of the model is the payoff for the contracting agents, which is discussed in the next section.

4. Wage Contracting Costs and Benefits

FISCHER [1977, 195] assumes that agents attempt to set the nominal wage "to maintain constancy of the real wage." Though problematical in the presence of supply shocks, this premise is consistent with labor-market agents remaining on their "offer curves" in the presence of demand shocks (AZARIADIS and COOPER [1985, 32]). Thus we can safely employ the premise here; but rather than assume a fixed money-wage contract, as Fischer does, we follow PHELPS' [1978, 209] suggestion that the choice of a particular wage-contracting strategy must depend on the policy that agents expect the authority to pursue. The determination of an optimal contracting strategy requires an analysis of the costs and benefits of contracting. Contracting costs include: (1) costs of negotiating a contract; (2) costs of monitoring the economy; and (3) costs to agents from having real wages diverge from their desired value.

In GRAY'S [1978] model, contracts entail fixed negotiation costs, an assumption we adopt as well. Following MCCALLUM [1983], a longer contract is assumed to involve lower negotiating costs (n) per time period. For contract length t, average per-period negotiating costs are n/t. We treat monitoring costs as a constant function of time and also as a function of the expected variance of economic disturbances (and, hence, of the expected monetary policy).

Also following GRAY'S [1978] approach, we specify the loss function for real wage deviations as:

$$d = 1/t\,(\lambda \int_o^t \phi_x dx)\,, \tag{8}$$

where

$$\phi_x = E\{[w_x - p_x - (w_o - p_o)]^2\}\,. \tag{9}$$

As MCCALLUM [1983, 415] notes, forecasts become less accurate and the risk of real wage deviations increases as contract length increases. In calculating contract length, we posit that the expected deviation loss is an increasing function of contract length. Deviations depend on policy, being least severe with an accommodative policy and more severe with a rigid or manipulative policy. The model calculates actual as well as expected deviation costs as a quadratic function of w and p (see section 5 and the Appendix for details).

The per-period utility U of a contract of length t equals the benefits minus the costs:

$$U(t) = b(t) - c_m - d(t) - n/t\,, \tag{10}$$

where $d(t)$ is $d(w_t - p_t)$ for a given time period, c_m is the monitoring costs, and n/t is the negotiating cost, as discussed above.

The benefits $b(t)$ of contracts are, following FLANAGAN [1984], assumed to include (1) risk-reduction: employees, being risk-averse, reduce the risks of wage reductions; (2) human capital preservation: employers decrease the costs of turnover and loss of training investment; and (3) principal-agent cost reduction: contracts provide performance incentives. We also assume that the gross benefits of fixed money-wage contracts are, like contract costs, a function of contract length, but that the marginal benefit of a longer contract diminishes after some optimal length as contract length increases, i.e. $\partial b/\partial t < 0$. If $\partial b/\partial t$ were always positive, contract lengths would be infinite. The reason that the benefit eventually decreases is that, as noted above, fixing a wage for too long a time period leaves the contracting parties vulnerable to uncertainty, which increases over time, e.g. to real or monetary changes in the external environment which may be unfavorable to either the workers or the owners. These could include changes in the price level and in the demand for labor, and possible changes in monetary policy. Such changes in the economic environment could exceed the value of avoiding real-wage reductions.

Since some costs of human capital, such as training, are fixed, the marginal benefits from the training diminish with time as the cost is spread over more years. The value of wage flexibility to employers would increase with increasing time, eventually becoming greater than the value of maintaining human capital. Note that if $\partial b/\partial t$ is linear (a constant), then it has to be either negative or smaller than $\partial d/\partial t$ in equation (12) below.

To maximize U, set the first derivative equal to zero:

$$\partial U/\partial t = \partial b/\partial t - \partial d/\partial t + n/(t^2) = 0. \tag{11}$$

This requires

$$\partial b/\partial t = \partial d/\partial t - n/(t^2)\,, \tag{12}$$

for some contract length t^*. If $b(t)$ and $d(t)$ are linear functions with first derivatives b' and d', $b' = d' - n/(t^2)$, and $t^* = \sqrt{n/(d' - b')}$. Not surprisingly, the optimal contract length t^* varies directly with marginal contracting benefits b and negotiating cost n, and inversely with marginal real-wage deviation costs d. In the explicit model, to obtain the contract length, the expected marginal benefit and costs must be computed for sequential time periods until the decreasing marginal benefit is less than the increasing marginal costs for the next time period. For variable-wage contracts, since the nominal wage is reset during each period, deviation costs are independent of policy and the contract lengths are the same.

This completes the model for a one shot game. A repeated game and a Stackelberg game are described following the analysis of one-shot outcomes.

5. *The Payoff Simulation*

In the explicit model and program, ρ, the autoregressive coefficient used in (3) for the next-period demand disturbance, is set to 0.5. As ρ gets closer to 1, the disturbances feed on themselves more strongly, making the price level move further from its original level over time. However, the program results are robust to various settings of ρ. The stochastic term η is generated by taking the average of five random numbers and subtracting 0.5, thus generating a number between -0.5 and $+0.5$, weighted towards zero. Taking an average of five numbers also reduces the variation in the results from different random sequences. The relative results are robust to the particular random sequence. Since the model contains the stochastic disturbance term δ, implemented with random numbers in the program, the exact numerical payoffs differ for each random-number sequence; relative game payoffs remain consistent. (See "*Random Variables*" in the Appendix).

In (4), the function g is the square of $k(y - y_r)$, with the same sign as $(y - y_r)$, where k is a parameter indicating the value to the authority of the deviation of y from what rigid policy would yield, G being positively related to positive deviations of y due to monetary policy. The function d is $g_d(v(w - p))^2$, where g_d and v are parameters; G is negatively related to unfavorable changes in the real wage due to price-level changes.

The cost c in (4) of implementing monetary policy is a function of the chosen policy. It is set in the program at zero for rigid, one for accommodative, and two for manipulative policy. (Because a manipulative policy first accommodates for demand disturbances and then adds its own intentional disturbance, it is assumed to be the most difficult and costly to implement of the three policies.)

For y we use equation (1). If the policy is rigid, $y_r = y$; otherwise y_t is computed as $(-\delta_t - w_t)$, the second term being that wage which would have been in effect with rigid policy (equal to the price level p that would have been in effect with $m = 0$, namely $-\delta_t$).

In (6), the explicit model implements the manipulative term as $\gamma(w + 0.5)$, where $\gamma = 1.5$, 0.5 being half the probability range of the demand disturbance.

For wage-contracting costs, monitoring costs are set for both variable and fixed-wage contracts to a benchmark value of 1 per period for a rigid monetary policy. If policy is accommodative, the monitoring cost is 1 for variable and 0 for fixed contracts. For manipulative policy, we set the cost as double that of the rigid case, since agents need to monitor the policy as well as the economy, and the expected variance is greater. For the monetary authority, monitoring costs are set at zero for rigid policy, 1 for accommodative policy, and 2 for manipulative policy, both parties having the same costs of accessing and processing information. While the government may have a greater capacity to process information, some of that data is available to the agents.

In (10), the benefit $b(t)$ in each contract period is assumed to be $s/z(t)$, where s is a positive parameter, $\partial z/\partial t > 0$, and $\partial^2 z/\partial z^2 < 0$; explicitly, this is implemented with $z(t) = 1 + \log(t)$.

To summarize the sequence taken by the program, first the policy option and contract type (fixed or variable wage) is input. The program computes the monitoring costs and contract length. Then it iterates over 20 periods, computing the following in sequence: random numbers for δ_t, the money supply, price level, the actual and rigid-policy wage (at the first period of a contract or each period for variable wages), the contract benefit for that period, the actual and rigid-policy outputs, the deviation cost, the payoff for the monetary authority, and the contracting agents' costs. After the iterations, the average payoff per period for the contracting agents and the monetary authority is computed and printed.

6. *Wage Contracting Payoffs*

The three monetary-policy options identified in section 3 and the two types of nominal-wage contracts, fixed or variable, imply six possible policy-contract combinations and associated gaming payoffs. Given some set of explicit parameter values, such as those used in the Appendix, and the option chosen by the other player, both contracting agents and the monetary authority play a pure strategy: the given information leads to a single payoff rather than any probability distribution among strategy choices.

The following describes a one shot game in which the payoffs do not take into account the actions of the other player. Subsequently, a repeated game is considered, in which the players learn to react to game plays. If the monetary authority is interpreted as a Stackelberg leader, the order of play being that the monetary authority chooses a policy action, and then the contracting agents choose a contracting action, an equilibrium result is also obtained. The information set is the same for both players, and includes the monetary rule.

So we have the following six cases.

6.1 Policy is Rigid; Contracts Allow for a Variable Wage

Agents expect economic disturbances to occur, and they incur a moderate cost of monitoring the economy to adjust the nominal wage each period and maintain a constant real wage. In the model, the cost d of incurred deviations of the real wage from its desired level is zero, since the nominal wage is adjusted whenever the real wage is found to deviate from what the contract specifies. Contract length is calculated at 10 periods, a relatively long time, as net marginal benefits, posited to be high at the beginning of a contract, decline as uncertainty increases over time. The contracting agents' computed payoff is 2.9 (all results are rounded here to the nearest tenth), which serves as a benchmark for the other payoff results.

The computed payoff and its value relative to the other cases depend on and are sensitive to the parameters, which are calibrated to produce the relative payoffs. These parameters embody the theory and assumptions of the model – for example, the greater monitoring costs of manipulative policy. The advantage of the quantitative model and program is in embodying the assumptions of the model *in the initial parameters* rather than implicitly implementing them by assigning arbitrary relative values for the game payoffs. The program allows the outcomes to reflect not just the parameters but also the equations of the model, i.e. the determination of wages, money, and the price level, since the parameters and equations are common to all payoffs.

From equation (4), the monetary authority gains nothing either through affecting income or by achieving other goals (u), but also has a cost of zero (ignoring of course possible opportunity costs of rigid policy). The overall payoff to the authority is therefore zero.

6.2 Policy is Rigid; Contracts Involve a Fixed Wage

With nominal wages fixed, deviation costs d will be incurred, since during the contract, money-demand shocks will make the real wage different form that which agents desired at the beginning of the contract. In a world with high uncertainty and a history of economic shocks, the expectation would be that d would be significant. The contract length is computed as 5. If agents adopt fixed-wage contracts and are faced with shocks greater than expected it would, as JACOBY and MITCHELL [1984, 219] observe, be "quite feasible for the parties to rip up existing contracts," as post-1979 experience has shown.

Under a rigid (money-targeting) policy, given significant demand shocks, the costs of contracts exceed those of a flexible-wage contract. The payoff to the contracting agents is computed as 2.1, less than that of case (1). The payoff to the monetary authority is computed as -0.17 (the particular number is sensitive to the random sequence, but is generally a small negative number). The reduced social welfare to the agents in the economy due to d has a mild adverse political effect, and the policy has the opportunity cost of not implementing a policy which would generate greater utility to the authority. Hence under a rigid policy, the variable-wage strategy dominates.

6.3 Policy is Accommodative; Contracts Allow for a Variable Wage

With monetary policy maintaining a stable price level, deviation costs are lower than in case (1). Just as with variable-wage contracts, the optimal contract length is computed as 10 periods, hence negotiating costs per period are lower. This agrees with MCCALLUM's [1983, 415] result that an accommodative policy increases contract length. The payoff to agents is 2.9, equal to that of case (1).

The monetary authority now enjoys a positive gain from preventing y from fluctuating due to changes in money demand, if such policy is effective. It also incurs a cost of monitoring economic variables, but no wage deviation cost d. Its net payoff is put at 1.

6.4 Policy is Accommodative; Contracts Involve a Fixed Wage

With a monitoring cost of zero and lower deviation costs, the cost of a fixed-wage contract is lower than for case (2). The cost is also lower than for the variable-wage strategy (3), since by premise any mild deviation cost of (4) is less than the elimination of the monitoring cost. The contracting agents' payoff is derived as 3.7, greater than that of case (3). Note that the deviation cost is reduced more for values of ρ closer to 1, since the unanticipated portion of the shocks becomes smaller. For example, if $\rho = 0.9$, the payoff for case (4) is computed to be 3.8, and if ρ is 0.95, the payoff is 3.9.

The worst that can happen if ρ is close to zero is that the money supply moves closer to a constant value, as with rigid policy, but the payoff is still lower than for case (2) due to the absence of the monitoring cost and also the lower contract-negotiating cost. If ρ is low, however, there is little accommodation taking place, and in reality the monitoring cost would be similar to that of the rigid case, so this case presumes that a significant portion of the demand shocks are autoregressive.

The payoff to the monetary authority from stabilizing y is computed as 2.2, greater than in case (3). It should be noted, however, that the mathematical expectation of the authority's payoff is 2, since the gain or loss from affecting output relative to the rigid case output can be positive or negative. The utility unrelated to output is set at 2 (versus 1 if contracts have variable wages) because the political payoff is greater. The computed payoff to the monetary authority is sensitive to the autoregression coefficient ρ, higher values of ρ generate values for money supply and the price level that are ever more removed from the rigid case, the result being sensitive to the initial disturbance. We presume, therefore, that ρ is not close to 1, leaving some significant scope to unanticipated disturbances. (It would also be unrealistic for the disturbances to continue to follow equation (3) with a fixed ρ indefinitely; ρ itself would be stochastic or follow some boom-bust sequence.)

The payoff to the monetary authority is greater than with rigid policy unless the unaccommodated shocks over a lengthy time period, with the same fixed-wage contract length, would have made output much greater; this would decrease the government's accommodation payoff since the actual output becomes less than in the rigid case. This is unlikely over many periods, and since the magnitude and sign of the shocks is unknown *a priori*, the expected average gain or loss from altering y is zero.

6.5 Policy is Manipulative; Contracts Allow for a Variable Wage

With the monetary authority pursuing goals beyond simply responding to shocks, agents' monitoring costs increase even more than in case (5), with lags in changing the nominal wage being more costly as well. With a contract length of 10, the payoff is calculated as 1.9. Fethke and Policano [1987, 90] note that agents will attempt to set their negotiating dates to take advantage of discretionary policy. Agents can minimize their costs by such measures, but the costs will still exceed those in cases (1) and (3).

Since money wages can vary, there is no anticipated gain to the monetary authority from a manipulative policy. The calculated payoff is -2, less than that of a zero-cost rigid policy.

6.6 *Policy is Manipulative; Contracts Involve a Fixed Wage*

The deviation cost for the contracting agents is higher than in the case of a rigid policy, so the agents incur a higher cost than in case (2). The optimal contract is 3 periods, so there is also a greater average negotiating cost. The computed payoff is -5.6.

The policymaker gains from making y greater. The program calculates the net gain to the monetary authority as 4.8. In this case, the greater political payoff comes from "unnatural" gains in output rather than from stabilization services.

A summary of the outcomes for each of the six cases is given in table 1.

Table 1

	Case					
	(1)	(2)	(3)	(4)	(5)	(6)
Policy	rigid	rigid	accom	accom	manip	manip
Contract	var	fixed	var	fixed	var	fixed
Length	10	5	10	10	10	3
Authority	0	− 0.17	1	2.2	−2	4.8
Agents	2.9	2.1	2.9	3.7	1.9	−5.6

Key: *length:* contract length in periods; *authority:* average per-period payoff for the monetary authority; *agents:* average per-period payoff for contracting agents.

The cases are summarized by the following payoff matrix:

Table 2

Monetary Authority

rigid *r*	accommodative *a*	manipulative *m*	Contracting Agents
0., 2.9	1, 2.9	−2, 1.9	*v* variable-wage contract
−0.17, 2.1	2.2, 3.7	4.8, −5.6	*f* fixed-wage contract

Key: The number to the left in each box is the monetary authority's payoff, and the number to the right is the payoff to contracting agents.

7. One-Shot Wage Contracting Strategies

If agents were at (f, r), a switch to variable-wage contracts would reduce the deviation cost, given a world with disturbances, resulting in a payoff gain. Agents would move to and remain in box (v, r), which dominates (f, r).

The monetary authority can in turn increase its payoff by moving from (v, r) to (v, a), adopting an accommodative policy. Though the cost of this policy is greater than in the rigid case, the authority benefits from being able to sell its stabilization services. The agents are also made better off through reduced deviation costs; but since they must still monitor the economy to periodically adjust the nominal wage, they now find it worthwhile to move to fixed-wage contracts, where monitoring costs are zero, achieving a higher net payoff to themselves in box (f, a). Accommodative policy dominates rigid policy for the monetary authority in either contracting option, and if the only action combinations were with rigid and accommodative policy, the equilibrium would be accommodative policy with fixed-wage contracts.[3]

Our response to Fischer could end here, having demonstrated that fixed-wage contracts may be a gaming response to accommodative policy. However, the model is applied more comprehensively to include manipulative policy, showing how such a policy is inconsistent with a non-cooperative Nash equilibrium. If the authorities attempt to exploit apparent rigidities, moving to (f, m), labor-market agents would respond by switching back to variable contracts, case (v, m). As FISCHER [1977, 204] himself remarks, "An attempt by the monetary authority to exploit the existing structure of contracts to produce behavior far different from that envisaged when contracts were signed would likely lead to the reopening of contracts." The payoff to the monetary authority then becomes negative, since it can no longer benefit by exploiting fixed-wage contracts, it incurs continuing monitoring costs, and it incurs negative reactions from the public due to social welfare losses. As FRIEDMAN [1977] notes, volatility reduces economic efficiency. This result is the counterpart of the result, stressed in the "time-consistency" literature, whereby the expectation that the authorities will be tempted to manipulate real wages merely serves to give rise to a higher equilibrium rate of inflation. Since in a one-shot game the monetary authority's best response is in case (f, m) and the agents' best response is (f, a), there is no Nash equilibrium. Manipulative action will lead to the outcome (v, m), but accommodative action will leave the monetary authority with a lower payoff than the manipulative policy. A one-shot game outcome is the cycle (f, a), (f, m), (v, m), (v, a), (f, a), and so on.

[3] Interestingly, our hypothetical example suggests that the monetary authority would never commit itself to a rigid, money-targeting policy since commitment to a price-targeting policy results in a much higher equilibrium payoff. This result seems in accord with the revealed preferences of real-world central bankers.

8. A Repeated-Game and Stackelberg Solution

If neither party is regarded as a Stackelberg leader, cycling generates a repeated game in which the players remember all past information and condition their play on past experience. Players seek to maximize the discounted expected value of the sequence of payoffs. At the beginning of the game, expectations are necessarily adaptive: future moves are uncertain, as each party, moving in a fog, has an unclear model of the other's behavior. As the game iterates, the contracting agents develop less uncertain expectations as to the monetary authority's payoffs; expectations become forward-looking and "rational" rather than adaptive. A well-defined equilibrium condition then exists, in which that players receive at least their minimax payoff (FUDENBERG and TIROLE [1991, 151]). In our case, the repeated game settles at (f, a), because this outcome provides long-term minimax gains to the monetary authority. This outcome resembles that of a cooperative game, but is in fact enforced by an infinite regress of threatened punishments.

Empirical evidence seems to bear out these conclusions. According to MCCALLUM [1983, 415], in the late 1950s the number of contracts with predetermined annual wage increases rose from 20% to 70%, possibly due to "*ex ante* perceptions of stabilization policy." Contract lengths also responded to changing conditions. A subsequent shortening of contract duration, according to CULLEN [1985, 307], reflected "a logical reaction by many negotiators to the turmoil and uncertainty of the early 1980s."

Cycling is avoided in the first place if the monetary authority is a Stackelberg leader, choosing its move first, with contracting agents observing the authority's move before choosing their best response. The two types of asymmetry noted by POHJOLA [1986] apply here: 1) the leader is able to announce a strategy first and impose it on the others; 2) the leader is able to act before the follower in every stage of the game. Because player 1 moves first, it cannot condition its payoff on player 2's (FUDENBERG and TIROLE, [1991, 67]), thus obviating the Fischer scenario. If the monetary authority plays a manipulative game, the contracting agents adopt variable-wage contracts, thus yielding a negative payoff to the monetary authority. Given the reaction of the second player, the first will choose the highest payoff at (f, a), just as in the repeated-game outcome, and as illustrated below. FAXÉN'S [1957] early dynamic game formulation

Table 3
Stackelberg Payoffs: Monetary Authority Leading

r	a	m	
0., 2.9	n/a	−2, 1.9	v
n/a	2.2, 3.7	n/a	f

of government/private-sector interaction is also applicable. In Faxén's model, the firm's reaction pattern is derived from the model, based on anticipated policy. The implication is that policies should be based on a consistent long-term strategy.

9. Conclusions

Sheffrin [1983, 50] observes that "By setting their nominal wage in advance, the workers [in Fischer's model] are giving the authorities a chance to stabilize the economy by *destabilizing* their real wages." But this would only happen if expectations were irrational. In a rational-expectations world, workers know the monetary rule and adopt long-term fixed-wage contracts only if they expect the authorities to accommodate money demand. This view casts doubt upon the robustness of Fischer's conclusions, while ironically rescuing them from Barro's [1977] objection that Fischer-type contracts are inconsistent with rational behavior.

Long-term contracts with a fixed money wage are, as Sheffrin [1983, 2] puts it, a "strategy for easing the computational burden on decision makers." Rather than giving scope for an activist monetary policy, fixed-wage contracts in a rational-expectations setting would provide only an extremely limited role for monetary policy. Policy may merely serve to make such contracts feasible when they might otherwise be ill-advised.

Appendix

The qualitative model described above by equations (1) through (12) is turned here into an explicit quantitative model that can be used to compute the payoffs of the "wage contracting" game played above. A computer program in the BASIC language based on this model is available from the authors. The model and program yield the payoffs for the policymaker and contracting agents of particular policy and contract combinations for a one-shot game.

The initial conditions of the model include some economic assumptions, such as the diminishing marginal benefit of a contract, some calibrating parameters or scale factors, and some programming "flags" (e.g. policy). The following are the parameters and their values:

Parameters

policy: 0 = rigid, 1 = accommodative, 2 = manipulative;
manip = 1 for manipulative policy, otherwise 0;
contract: 0 = variable, 1 = fixed-wage;
active = 0 for rigid policy, 1 otherwise;
gamma = 1.4 (coefficient of manipulation);

j	= 2 for variable and 3 for fixed contracts (coefficient of policy benefit for accommodative policy);
$k = 1.5$	(coefficient of policy success);
$r = 3$	(coefficient of negotiating cost);
rho = .5	(coefficient of disturbance correlation);
$s = 10$	(coefficient of contract benefit);
$v = 1$	(coefficient of wage deviation);
gd = .8	(coefficient of government's deviating cost);
years = 20	(number of periods used in calculating payoffs).

Monitoring Cost per Year

Monitoring costs for the monetary authority are equal to the policy parameter. For the contracting agents, monitoring costs are 1 for rigid policy and for accommodative policy with variable-wage contracts, 0 for accommodative policy with fixed-wage contracts, and 2 for manipulative policy.

Contract Length

Find $\partial b/\partial t = \partial d/\partial t - n/t^2$, solving for t.

Contract length is calculated by iterating over contract periods until the marginal benefit of another period is less than the marginal cost. For each iteration, the expected deviation *wx* is calculated from the initial real wage level of zero. The deviations increase as the number of periods *lt* increases. Equation (9) is implemented in the program by assigning a larger weight for manipulative policy and a lower weight for accommodative policy. The contract benefit is calculated as a ratio, with the numerator the parameter *s* and the denominator equal to $(1 + \log(lt))$. The values are calibrated to produce a reasonable benefit schedule. The algorithm then compares the differences in benefits and costs between the current and past iteration, per equation (12). For variable-length contracts, deviation costs are treated as identical to that under accommodative policy.

Random Variables

RND is a random number between 0 and 1, generated by the computer. The variable *rand* = (RND + RND + RND + RND + RND)/5 − 0.5 to center the shock variable around zero and reduce the likelihood of values at the extremes, which could perpetuate an increasing or decreasing accumulated disturbance sum with the autoregressive scheme, $delta_t = rho*delta_{t-1} + rand$, in accord with equation (3), where *delta* is the disturbance term for the model. The *deltas* are accumulated in the variable *tdelta*.

Money Supply Change

The money supply *m* is zero for rigid policy. For accommodative policy, $m = rho*tdelta$, in accord with equation (5). Manipulative policy first sets

m as in the accommodative case, and then adds a manipulative element: $m = m + gamma* (w + 0.5)$, since its object is to increase the price level relative to wages.

Price Level

$p = m - tdelta.$

As specified in equation (7), the price level is a function of the money supply as well as past demand shocks.

Wage Level Setting

The nominal wage is w; $w = p$ at the beginning of a contract and, for flexible contracts, at each contract period. Otherwise, w remains constant during the duration of a contract for fixed-wage contracts.

Output

$y = p - w;\ y_r = -\delta_p - w_r.$

For y in equation (4), actual output (y in the program) is set at the price level minus the wage level. Also computed is y_r, the output with rigid policy, computed as the negative of the accumulated deltas (which would be equal to the price level with m at zero) less the wage that would be set with rigid policy, set at the same time as w, equal to the negative of *tdelta* (hence y_r, is zero for flexible wages and in the first period of fixed-wage contracts).

Actual Deviation Loss

$d = (v(w - p))^2.$

The cost of the deviation of the wage from the price level is a quadratic function of wage w and price level p, v indicating the severity of the loss.

Policymaker Payoff

$G = \text{sign}(y - y_r)*(k*(y - y_r))^2 - d*gd + j*active*(1 - manip) - c_m.$

Implementing equation (4), the payoff to the policymaker is a quadratic function of $y - y_r$ (according to the sign of the difference), the agents' deviation loss d, the utility gain (u) from parameters j and gd, and the monitoring cost c_m. The payoff depends on parameter k, the "coefficient of policy success." A high k gives manipulative policy a higher payoff to the monetary authority. The payoff is computed as the average per period over 20 periods.

Contract Benefit

Benefit $= s/(1 + \log(cl))$, where cl is the current period of the contract. The contract benefit is calculated the same way as in the calculation of the contract length.

Contract Costs

$c_t = c_{ma} + d.$

Once the contract length is set, the variable annual contracting costs are made up of the agents' monitoring cost (c_{ma}) and the deviation cost.

Contracting Agents' Payoff

$\text{Payoff} = \text{Benefit} - c_{ma} - d - \text{years}^* r/(\text{contract length})^2.$

Per equation (10), the net payoff for the contracting agents equals the benefit minus the monitoring cost minus the weighted deviation cost minus the contracting cost. The payoff is computed as the average per period.

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